

1 Basic Knowledge & FPM

1.1 Definitions, Useful Tools & Nested Intervals

Def of $\lambda(I)$: $\lambda((a, b)) = \lambda([a, b)) = \lambda((a, b]) = \lambda([a, b]) = b - a$.

· **Def of $\chi_I(x)$:** $\chi_I(x) := \mathbb{1}_{x \in I}$

Def of limit superior and inferior: $\limsup a_n := \lim(\sup_{k \geq n} a_k)$

$\liminf a_n := \lim(\inf_{k \geq n} a_k)$

Important Inequation: $|x + y| \leq |x| + |y|$; $|x - y| \geq |x| - |y|$; $||x| - |y|| \leq |x \pm y|$. $|\sin(x)| \leq |x|$ ps: need prove, By MVT

Bernoulli's Inequality: If $\alpha > 0, \delta \geq -1$. Then $(1 + \delta)^\alpha \geq 1 + \delta\alpha, \alpha \in [1, +\infty)$

Def of Nested Interval: Sequence $(I_n)_{n \in \mathbb{N}}$ is nested if $I_n \supseteq I_{n+1}$ for all $n \in \mathbb{N}$. · **MCT:** $UB \Rightarrow \sup; LB \Rightarrow \inf$.

Nested Interval Theorem: If $(I_n)_{n \in \mathbb{N}}$ is nested closed bounded interval sequence $\Rightarrow E := \cap_n I_n \neq \emptyset$. If $\lim \lambda(I_n) = 0 \Rightarrow E = a, a \in \mathbb{R}$

Power Set $\mathcal{P}$: $\mathcal{P}(E)$ is the set of all subsets of E . ps: $\mathcal{P}(\mathbb{R})$ is the set of all subsets of $\mathbb{R}$.

1.2 Infinite Sequence, Cauchy Sequence and Bolzano-Weierstrass Theorem

Useful Seq Limit: $\lim \frac{1}{n^p} = 0, p > 1$; $\lim a^n = 0, |a| < 1$; $\lim n^{\frac{1}{n}} = 1$; $\lim a^{\frac{1}{n}} = 1, a > 0$

Subsequence Theorem: In each case, the subsequence can be taken to be monotonic.

1. $\exists a \in \mathbb{R}$ s.t. $\forall \varepsilon > 0$, there are infinitely many $n \in \mathbb{N}$ s.t. $|x_n - a| < \varepsilon \Leftrightarrow \exists$ subsequence of (x_n) s.t. $\lim x_{n_k} = a$

2. $(x_n)_{n \in \mathbb{N}}$ is not bounded above (below) $\Leftrightarrow \exists$ subsequence of x_n s.t. $\lim x_{n_k} = \infty (-\infty)$

3. If $\lim x_n = L \Rightarrow \forall x_{n_k}, \lim x_{n_k} = L$

Cauchy Sequence: Sequence $(a_n)_{n \in \mathbb{N}}$ is Cauchy iff $\forall \varepsilon > 0, \exists N \in \mathbb{N}$ s.t. $|a_n - a_m| < \varepsilon$ for all $m, n \geq N$. · **Convergent $\Leftrightarrow$ Cauchy**

Bolzano-Weierstrass Theorem: Every bounded sequence has a convergent subsequence. ps: In real number.

1.3 Infinite Series and their theorems

· **p-Test:** $\sum \frac{1}{n^p}$ convergent iff $p > 1$ **Root Test** **Ratio Test** $|a_n|^{1/n}, |a_{n+1}/a_n|$ **Geometric Series:** $\sum p^n$ converges iff $|p| < 1$

Divergence Test: If $\lim a_n \neq 0 \Rightarrow \sum a_n$ diverges. (If $\sum a_n$ convergent $\Rightarrow \lim a_n = 0$.)

Comparison Test: If $0 < a_n < b_n, \sum b_n$ convergent $\Rightarrow \sum a_n$ also; $\sum a_n$ divergent $\Rightarrow \sum b_n$ also.

· **Limit Comparison Test:** If $a_n, b_n \geq 0, \lim \frac{a_n}{b_n} = L > 0 \Rightarrow$ If $L \in (0, \infty)$: $\sum a_n$ convergent iff $\sum b_n$ also.

If $L = 0$: $\sum b_n$ convergent $\Rightarrow \sum a_n$ also. $L = \infty$: $\sum b_n$ divergent $\Rightarrow \sum a_n$ also.

Integral Test: Let $f: [1, \infty) \rightarrow \mathbb{R}$ is 非负递减, $a_n = f(n)$. Then $\sum a_n$ converges iff $\int_1^\infty f(x)dx < \infty$.

Absolutely Convergence: $\sum a_n$ convergent absolutely iff $\sum |a_n|$ convergent. **If convergent abs $\Rightarrow$ convergent.**

Alternating Series Test: If a_n decreasing, $a_n \geq 0, \lim a_n = 0$. Then $\sum (-1)^{n-1} a_n$ convergent.

Cauchy's Condensation Test: If $a_n \geq 0, a_n$ decreasing, $\Rightarrow [\sum a_n \text{ convergent} \Leftrightarrow \sum 2^n a_{2^n} \text{ also}]$

Cauchy Criterion for Series: $\sum_{k=1}^\infty a_k$ converges iff $\forall \varepsilon > 0, \exists N \in \mathbb{N}$ s.t. $\forall n \geq m \geq N, |\sum_{k=m+1}^n a_k| = |a_{m+1} + \dots + a_n| < \varepsilon$.

Riemann Rearrangement Theorem: If $\sum_{k=1}^\infty a_k$ is conditionally convergent. $\Rightarrow \forall s \in \mathbb{R}, \exists$ bijection $\sigma: \mathbb{N} \rightarrow \mathbb{N}$ s.t. $\sum_{n=1}^\infty a_{\sigma(n)} = s$.

Rearrangement for Abs Series: If $\sum_{n=1}^\infty a_n$ is absolutely convergent. $\Rightarrow \forall$ bijection $\sigma: \mathbb{N} \rightarrow \mathbb{N}, \sum_{n=1}^\infty a_{\sigma(n)} = \sum_{n=1}^\infty a_n$.

1.4 Continuous, differentiable functions and their theorems

Def of lim of func: $1. \forall \varepsilon > 0, \exists \delta > 0$ s.t. $0 < |x - x_0| < \delta \Rightarrow |f(x) - L| < \varepsilon$. $2. \forall x_n \neq x_0$ s.t. $\lim(x_n) = x_0$, then $\lim f(x_n) = L$.

Def of cont of func: $1. \forall \varepsilon > 0, \exists \delta > 0$ s.t. $\forall x, |x - x_0| < \delta \Rightarrow |f(x) - f(x_0)| < \varepsilon$. $2. \forall (x_n)_{n \in \mathbb{N}}, \text{if } \lim x_n = x_0 \Rightarrow \lim f(x_n) = f(x_0)$

Def of diff: $1. f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$ $2. f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$.

The Extreme Value Theorem: If $f: [a, b] \rightarrow \mathbb{R}$ is continuous, then f has a max and min value.

Intermediate Value Theorem: $f: [a, b] \rightarrow \mathbb{R}$ is continuo, $f(a) \neq f(b), y \in [f(a), f(b)] \Rightarrow \exists c \in (a, b)$ s.t. $f(c) = y$

IVT (for deriv) / (Darboux Theorem): If f differentiable in $I \Rightarrow \forall m \in (f'(a), f'(b))$ we have: $\exists c$ s.t. $f'(c) = m$

Rolle's Theorem: If f continuous on $[a, b]$, differentiable on (a, b) and $f(a) = f(b)$ Then: $\exists c \in (a, b)$ s.t. $f'(c) = 0$

Mean Value Theorem: Let f, g continuous on $[a, b]$ and differentiable on (a, b) : ps: a=b 时依然成立

· $1. \exists c \in (a, b)$ s.t. $\frac{f(b) - f(a)}{b - a} = f'(c)$ $2. \exists c \in (a, b)$ s.t. $\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(c)}{g'(c)}$

Theorem: If f is 1-1 and continuous $\Rightarrow f$ is strictly monotone. and f^{-1} is continuous and strictly monotone.

Inverse Function Theorem: Let f differentiable in I (open interval), and $\forall x, f'(x) \neq 0$:

· $1. f(I)$ is open interval. $2. f$ is invertible. $3. f^{-1}$ is differentiable and $(f^{-1})'(y) = \frac{1}{f'(f^{-1}(y))}$

Second Deriv Test: f twice continuously differentiable in $I, f'(x_0) = 0$: $1. f''(x_0) > 0 (< 0)$ local min (max) point.

2 Uniformly & Pointwise Convergence

2.1 Pointwise Convergence

Def of Pointwise Convergence: $1. f_n: E \rightarrow \mathbb{R}$ converge pointwise iff $\forall x \in E, \forall \varepsilon > 0, \exists N_{\varepsilon, x}$ s.t. $\forall n > N, |f_n(x) - f(x)| < \varepsilon$ $2. \forall x \in E, \lim f_n(x)$ exists $3. f(x) = \lim f_n(x)$

· pointwise convergence do NOT preserve: continuity, differentiability, $(\lim f_n)' \neq \lim f_n', \int \lim f_n \neq \lim \int f_n$

2.2 Uniformly Convergence

Def of Graph: $\Gamma(f) := \{(x, f(x)) | x \in E\}$ ps: $f: E \rightarrow \mathbb{R}$

Def of Vertical Neighbourhood $\mathcal{N}(f, \varepsilon) := \{(x, y) | x \in E, |y - f(x)| < \varepsilon\}$ ps: $f: E \rightarrow \mathbb{R}$

Def of Uniformly Convergence: $f_n: E \rightarrow \mathbb{R}$ converge uniformly iff $\forall \varepsilon > 0, \exists N_\varepsilon$ s.t. $\forall n > N, \forall x \in E, |f_n(x) - f(x)| < \varepsilon$ Equivalent to

1. $\lim [\sup_{x \in E} |f_n(x) - f(x)|] = 0$ · **Uniformly $\Rightarrow$ Pointwise**

2. **Vertical Neighbourhood Test:** $\forall \varepsilon > 0, \exists N \in \mathbb{N}$ s.t. $\forall n > N, \Gamma(f_n) \subset \mathcal{N}(f, \varepsilon)$

Uniformly Cauchy: $\forall \varepsilon > 0, \exists N \in \mathbb{N}$ s.t. $\forall x \in E, m, n > N |f_n(x) - f_m(x)| < \varepsilon$ ps: $f_n: E \rightarrow \mathbb{R}$ · **Converge uniformly $\Leftrightarrow$ Uniformly Cauchy**

Sequence Test: If $f_n: E \rightarrow \mathbb{R}$ converge uniformly to f . $\Rightarrow \forall (x_n)_{n \in \mathbb{N}} \in E, \lim |f_n(x_n) - f(x_n)| = 0$ 测试不是时可以用

Property of Uniformly Convergent: Let $f_n, f, g: I \rightarrow \mathbb{R}$ Then: NOT preserve: differentiable. ps: 2. Termwise differentiation

1. If $\lim f_n(x) = f(x)$ uniformly, $\forall f_n$ continuous at $x_0 \Rightarrow f$ continuous at x_0 . $\Downarrow \forall x \in I, (\lim f_n)'(x) = \lim f_n'(x)$

2. If I is bounded open; $\forall f_n$ diff, $\lim f_n' = g$ unif; $\exists x_0 \in I$ s.t. $f_n(x_0)$ converges. $\Rightarrow \lim f_n = f$ unif; f' diff and $f' = g$.

3. **Cannot use now** If $\lim f_n(x) = f(x)$ uniformly on closed interval $[a, b]$, $\forall f_n$ integrable on $[a, b] \Rightarrow \lim \int_a^b f_n(x) dx = \int_a^b \lim f_n(x) dx$

Def of Series of Functions: Let $f_n: E \rightarrow \mathbb{R}$; Def partial sum $s_n(x) := \sum_{k=1}^n f_k(x)$

1. $\sum f_n$ converge pointwise iff s_n converge pointwise. 2. $\sum f_n$ converge uniformly iff s_n converge uniformly. 3. $\sum f_n$ converge absolutely (pointwise) iff $\sum |f_n|$ converge (pointwise).

Weierstrass M-Test: $f_n: E \rightarrow \mathbb{R}, \exists (M_n)_{n \in \mathbb{N}} \geq 0$ s.t. $\forall x \in E, \forall n \in \mathbb{N}: |f_n| < M_n$; $\sum M_n$ converges. $\Rightarrow \sum f_n$ converge absolutely & Uniformly.

3 Power Series

3.1 Introduction, Radius of Convergence, Continuity and Differentiability of Power Series

Power Series: A series from $\sum_{n=0}^{\infty} a_n(x-c)^n$ **Center:** c . **Coefficients:** a_n .

Radius of Convergence (R): For a power series $\sum a_n(x-c)^n$, there are three situations for it's radius of convergence:

1. $R = 0$ if $\sum a_n(x-c)^n$ converges only $x = c$ || $R = \infty$ if $\sum a_n(x-c)^n$ converges absolutely $\forall x \in \mathbb{R}$

2. $\exists R > 0$ s.t. $\sum a_n(x-c)^n$ converges absolutely if $|x-c| < R$; diverges if $|x-c| > R$. || $R = \sup\{r \geq 0 : (a_n r^n)_{n \in \mathbb{N}} \text{ bounded}\}$

Theorem (Continuity, Uniformly and Differentiability): If $f(x) = \sum a_n(x-c)^n$ with $R > 0$.

1. $f(x)$ continuous on $(c-R, c+R)$

2. $\forall 0 < r < R, f(x)$ converges absolutely and uniformly on $[c-r, c+r]$

3. $\sum_{n=0}^{\infty} a_n(x-c)^n$ and $\sum_{n=1}^{\infty} n a_n(x-c)^{n-1}$ have same R .

4. **Termwise differentiation:** $f(x)$ is differentiable on $(c-R, c+R)$ || $f'(x) = \sum_{n=1}^{\infty} n a_n(x-c)^{n-1}, x \in (c-R, c+R)$
 $\forall 0 < r < R, f(x), f'(x)$ converges absolutely and uniformly on $[c-r, c+r]$

5. $f(x)$ is infinitely convergence on $(c-R, c+R)$ || $a_k = \frac{f^{(k)}(c)}{k!}$

3.2 The Exponential Function & Real Analytic Function

The Exponential Function: $E(x) := \sum_{n=0}^{\infty} \frac{x^n}{n!}$ with $R = \infty$ ps: usually denoted by $\exp(x)$ or e^x

Real Analytic Function: $f: I \rightarrow \mathbb{R}$ is real analytic iff $\exists (a_n)_{n \in \mathbb{N}}$ s.t. $f(x) = \sum_{n=0}^{\infty} a_n(x-c)^n$ for $\forall x \in I$ i.e. Series converges pointwise on I .

Taylor Series: Given 1 Open Interval I , 2 infinitely differentiable $f: I \rightarrow \mathbb{R}$. $\Rightarrow$ Taylor Series of f at $c \in I := \sum_{n=0}^{\infty} \frac{f^{(n)}(c)}{n!} (x-c)^n$

· Not all infinitely differentiable functions are real analytic: e.g. $f(x) = \{e^{-1/x^2}, x > 0; 0, x < 0\}$

Taylor's Theorem: If $f: (a, b) \rightarrow \mathbb{R}$ is differentiable function at $x_0 \in (a, b)$:

1. $P_{n,f,x_0}(x) := f(x_0) + \sum_{k=1}^n \frac{f^{(k)}(x_0)}{k!} (x-x_0)^k$ (Taylor's polynomial of degree n at x_0)

2. If $f^{(n+1)}$ exists on (a, b) , then: $\exists \xi \in (x, x_0)$ s.t. $f(x) = P_{n,f,x_0}(x) + \frac{f^{(n+1)}(\xi)}{(n+1)!} (x-x_0)^{n+1}$

3. Remainder $R_{n,f,x_0} := f(x) - P_{n,f,x_0}$. || $f: I \rightarrow \mathbb{R}$ is real analytic iff $\lim R_{n,f,x_0} = 0$ for $\forall x \in I$

4 Integration Theory & Lebesgue Integration

4.1 Introduction

4.1.1 Definition of Integral and Step Function

Types of Integration & Fundamental Theorem of Calculus:

1. **Indefinite integrals:** Also called *primitives* or *anti-derivatives*. ps: 不定积分. || Definition: 求导的逆运算 || e.g. $\int f'(x) dx = f(x) + c$

2. **Definite integrals:** ps: 定积分 || Definition: 函数在指定区域下的面积 || $\int_a^b f'(x) dx = f(b) - f(a)$

3. **Fundamental Theorem of Calculus:** $\int_a^b f'(x) dx = f(b) - f(a)$

Characteristic function χ_E : Given $E \subset \mathbb{R}, \chi_E(x) := \{1, x \in E; 0, x \notin E\}$. || ps: $\lambda(I) = \sup I - \inf I$ | If $I = I_1 \cup I_2, I_1 \cap I_2 = \emptyset \Rightarrow \lambda(I) = \lambda(I_1) + \lambda(I_2)$

Def of Simple Integral: 1. $\int_{\mathbb{R}} \chi_I(x) dx := \lambda(I)$ 2. $\int_{\mathbb{R}} c \chi_I(x) dx := c \lambda(I)$ 3. $\int_{\mathbb{R}} \sum_j c_j \chi_{I_j}(x) dx = \sum_j c_j \lambda(I_j)$

Step Function: non-negative $f: \mathbb{R} \rightarrow [0, \infty)$ is a step function iff $f(x) = \sum_j c_j \chi_{I_j}(x)$ || Def: $\int_{\mathbb{R}} f(x) dx = \sum_j c_j \lambda(I_j)$

4.1.2 Open Set and Closed Set

Open Set: A set S is open iff $\forall x \in S, \exists r > 0, s.t. (x - r, x + r) \subseteq S$ ps: Collection of all open subsets of $\mathbb{R}$ is called *the Euclidean topology*

Structure of open sets: If S is a nonempty open set $\Leftrightarrow S = \text{countable union of disjoint open intervals}$. ps: 即 open set = $\cup$ open intervals

$\cup$ of open sets: Let $\mathcal{U}$ be collection of some open sets. (Can Infinite) $\cup_{S \in \mathcal{U}} S$ is open.

$\cap$ of open sets: Let $\mathcal{U}$ be collection of some open sets. (Finite) $\cap_{S \in \mathcal{U}} S$ is open.

Connection with continuous function|open set: If $f : \mathbb{R} \rightarrow \mathbb{R} \Rightarrow \{x \in \mathbb{R} : f(x) > \lambda\}$ is open.

Closed Set: A set $F \subseteq \mathbb{R}$ is *closed* iff $\mathbb{R} \setminus F$ is open.

Connection with continuous function|open set: If $f : \mathbb{R} \rightarrow \mathbb{R} \Rightarrow \{x \in \mathbb{R} : f(x) \leq \lambda\}$ is closed.

Theorem: If F is closed set, $(x_n)_{n \in \mathbb{N}} \in F$ convergent. $\Rightarrow x = \lim x_n \in F$

4.2 Lebesgue Measures

Countably Additivity: $(E_n)_{n \in \mathbb{N}}$ disjoint sets, m be measure. $\Rightarrow$ Countably Additive if: $m(\cup_n E_n) = \sum m(E_n)$

Lebesgue Outer Measure: $m^*(E) := \inf\{\sum_{n=1}^\infty \lambda(I_n) \mid E \subseteq \cup_{n=1}^\infty I_n, I_n \text{ is a open interval}\}$ ps: 本质是寻找最小的区间可以覆盖 E , 用前面 λ 的定义来定义“长度”

· **Lebesgue Outer Measure (Version 2):** $m^*(E) = \inf\{\lambda(U), E \subseteq U, U \text{ is open set}\}$ ps: $\lambda(U) = \sum \lambda(I_n)$

· **Remark:** If it's impossible to find $(I_n)_{n \in \mathbb{N}}$ s.t. $\sum \lambda(I_n) < \infty \Rightarrow m^*(E) = \infty$ || Can interpret m^* as $m^* : \mathcal{P}(\mathbb{R}) \rightarrow [0, \infty)$

5 Appendix

Trigonometric func: $s+s=sc, s-s=cs, c+c=cc, c-c=-ss$ || e.g. $\sin A + \sin B = 2\sin(\frac{A+B}{2})\cos(\frac{A-B}{2})$ $\sin A \sin B = -\frac{1}{2}[\cos(A+B) - \cos(A-B)]$

hyperbolic func: $\sinh(x) = \frac{e^x - e^{-x}}{2}$ $\cosh(x) = \frac{e^x + e^{-x}}{2}$ $\cosh^2(x) - \sinh^2(x) = 1$ $[\sinh(x)]' = \cosh(x)$ $[\cosh(x)]' = \sinh(x)$

Common Int: $\int \tan(x)dx = -\ln|\cos(x)|$; $\int \cot(x)dx = \ln|\sin(x)|$; $\int \sec(x)dx = \ln|\sec(x) + \tan(x)|$; $\int \csc(x)dx = \ln|\csc(x) - \cot(x)| = \ln|\tan(\frac{x}{2})|$

Common Der: $[\tan(x)]' = \sec^2(x)$; $[\cot(x)]' = -\csc^2(x)$; $[\sec(x)]' = \sec(x)\tan(x)$; $[\csc(x)]' = -\csc(x)\cot(x)$; $[\arctan(x)]' = \frac{1}{1+x^2}$.
· $1 + \tan^2(x) = \sec^2(x)$; $1 - \csc^2(x) = \cot^2(x)$; $\csc^2(x) - \cot^2(x) = 1$. $[\arcsin(x)]' = \frac{1}{\sqrt{1-x^2}}$; $[\arccos(x)]' = -\frac{1}{\sqrt{1-x^2}}$

$q^n - 1$: $q^n - 1 = (q - 1)(1 + q + q^2 + \dots + q^{n-1})$ || **$1 \pm x^3$:** $1 \pm x^3 = (1 \pm x)(1 \mp x + x^2)$

Summation By Parts: $\sum_{k=1}^n a_k(b_{k+1} - b_k) = (a_{n+1}b_{n+1} - a_1b_1) - \sum_{k=1}^n (a_{k+1} - a_k)b_{k+1}, \forall (a_k)_{k=1}^{n+1}, (b_k)_{k=1}^{n+1}$
 $\Rightarrow \sum_{k=1}^n a_k b_k = a_{n+1}b_{n+1} - \sum_{k=1}^n (a_{k+1} - a_k)\sigma_{k+1}, \forall (a_k)_{k=1}^{n+1}, (b_k)_{k=1}^{n+1} \sigma_k := \sum_{j=1}^{k-1} b_j$

Common Power Series: $\sum_{n=0}^\infty x^n = (1 - x)^{-1}$ $\sum_{n=1}^\infty nx^{n-1} = (1 - x)^{-2}$ $\sum_{n=2}^\infty n(n-1)x^{n-2} = 2(1 - x)^{-3}$ $R = 1$ for all.
 $\sum_{n=0}^\infty \frac{x^n}{n!} = e^x$ $\sum_{n=1}^\infty (-1)^{n+1} \frac{x^n}{n} = \ln(x + 1)$ $\sum_{n=1}^\infty \frac{x^n}{n} = -\ln(1 - x)$ $R = \infty$ for all.

Sum of n, n^2, n^3 : $\sum_{k=1}^n k = \frac{n(n+1)}{2}$ $\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$ $\sum_{k=1}^n k^3 = [\frac{n(n+1)}{2}]^2$

Useful Conterexample: $(1/2)^{\frac{1}{n}} \subseteq [1/2, 1)$